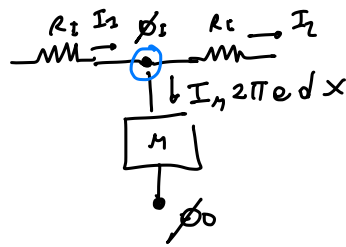


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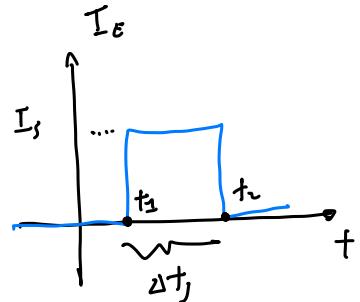
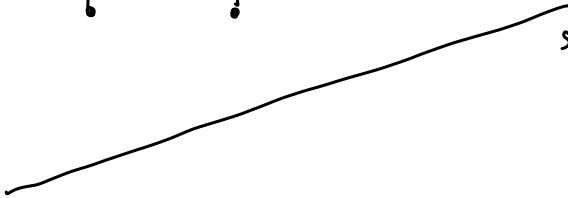
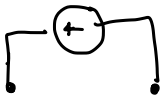
$$\frac{dV_n}{dt} = \frac{1}{C_n} \left[\frac{a G_L}{2} \frac{d^2 \phi_L}{ds^2} - I_{ion} \right]$$

$$V_n = \phi_L - \phi_0$$

$$\frac{dV_n(s,t)}{dt} = \frac{1}{C_n} \left[\frac{a G_L}{2} \frac{d^2 V_n(s,t)}{ds^2} + \frac{a G_L}{2} \frac{d^2 \phi_0(s,t)}{ds^2} - I_{ion} \right]$$

INPUT TERM!

$I_E(t)$



$$t < t_1 \rightarrow \frac{d^2 \phi_0}{ds^2} = 0 \quad \text{REST}$$

$$V_n = V_L \cong -70 \text{ mV}$$

$$\frac{dV_n}{dt} = 0$$

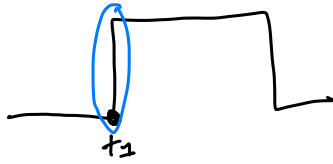
$$t_1 < t < t_2 \quad \frac{d^2 \phi_0}{ds^2} \neq 0 \rightarrow V_n \text{ CHANGES}$$

INCREASES

DECREASES

$$\frac{\partial^2 \phi_0}{\partial s^2} = F \quad \text{ACTIVATION FUNCTION} \longrightarrow F > 0 \longrightarrow \text{INCREASES IN } V_M$$

WHY?

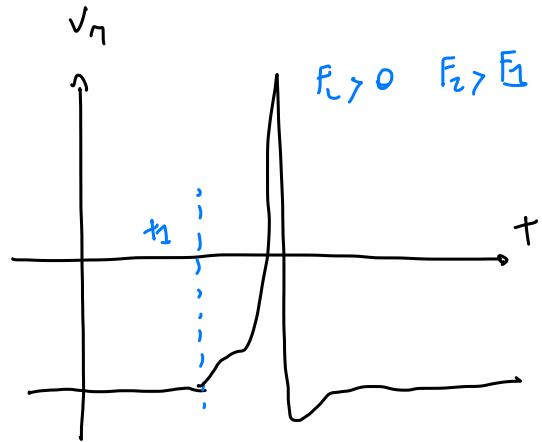
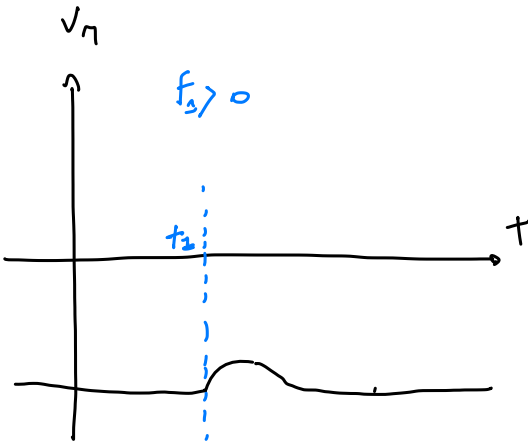


t_1^+
INPUT TERM

$$\frac{\partial V_M(s, t)}{\partial t} = \frac{1}{C_M} \left[\underbrace{\frac{e G_L}{2} \frac{\partial^2 V_M(s, t)}{\partial s^2}}_{t_1^+ \rightarrow \phi} + \frac{e G_L}{2} \boxed{\frac{\partial^2 \phi_0(s, t)}{\partial s^2}} - \underbrace{I_{ion}}_{t_1^+ I_{ion} = 0} \right]$$

$t_1^+ > 0$

IF $F > 0 \Rightarrow$ INCREASE IN V_M



$F > 0$ NECESSARY CONDITION TO STIMULATION $\longrightarrow$ NOT SUFFICIENT $\begin{cases} \nearrow \text{STIMULUS TO LOW} \\ \searrow T = \text{HIGH} \dots \end{cases}$

LINE CURRENT
SOURCE



$i_M \rightarrow$ MEMBRANE CURR.
PER UNIT OF LENGTH $\left[\frac{\text{mA}}{\text{m}} \right]$

$$I_{MA} = I_M 2\pi a dx = i_M dx$$

$$i_M = I_M 2\pi a$$

$$\cancel{e_e} \frac{\partial^2 V}{\partial t^2} + \cancel{e_e} \frac{\partial V}{\partial t} + \nabla \cdot \vec{F} = f \quad e_e = \phi$$

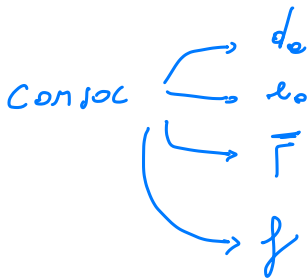
$$\boxed{\left(\frac{2 C_M}{a} \right) \frac{\partial V_M}{\partial t} + \frac{\partial^2}{\partial s^2} \left[-G_L V_M - G_C \phi_0 \right]} = \boxed{\left(-\frac{2}{e} \right) I_{ion}} \quad \begin{matrix} \downarrow f \\ \downarrow \end{matrix}$$

$$\frac{\partial}{\partial s} \left[-G_L \frac{\partial V_M}{\partial s} - G_C \frac{\partial \phi_0}{\partial s} \right] \quad \uparrow \vec{F}$$

$$\vec{F} = \boxed{-G_L \left(\frac{\partial V_M}{\partial s} + \frac{\partial \phi_0}{\partial s} \right) \hat{s}}$$

$$\nabla \cdot \vec{F} = \frac{d}{ds} \Gamma_s = \frac{d}{ds} \left[-G_I \left(\frac{dV_m}{ds} + \frac{d\phi_0}{ds} \right) \right] =$$

$$= \frac{d^2}{ds^2} \left[-G_I V_m - G_I \phi_0 \right]$$



$$\frac{dV}{ds} = \nabla V \cdot \hat{s} = \left(\frac{dV}{dx} \hat{x} + \frac{dV}{dy} \hat{y} + \frac{dV}{dz} \hat{z} \right) \cdot \hat{s} =$$

$$= \underbrace{\frac{dV}{dx} \hat{x} \cdot \hat{s}}_{UTX} + \underbrace{\frac{dV}{dy} \hat{y} \cdot \hat{s}}_{UTY} + \underbrace{\frac{dV}{dz} \hat{z} \cdot \hat{s}}_{UTZ} \rightarrow \begin{matrix} \text{quick} \\ \text{in} \\ \text{in consol} \end{matrix}$$

$$\frac{dV_m}{ds} = V_{mTx} + V_{mTy} + V_{mTz} = K_1$$

$$\frac{d\phi_0}{ds} = \phi_{0Tx} + \phi_{0Ty} + \phi_{0Tz} = K_2$$

$$\vec{F} = -G_I (K_1 + K_2) \hat{s} \quad \vec{F} = K_3 \hat{s}$$

$$\Gamma_x = K_3 \circlearrowleft s_x \rightarrow t_{1x}$$

$$\Gamma_y = K_3 \circlearrowleft s_y \rightarrow t_{1y}$$

$$\Gamma_z = K_3 \circlearrowleft s_z \rightarrow t_{1z}$$

INITIAL VALUES

$$V_A = V_R$$

$$\frac{dV_A}{dt} = 0$$

BOUNDARY
CONDITIONS



ZERO AT AXON
FLUX END)



QUIET IN
IN CONSOLE

$$U_2 = \begin{pmatrix} n \\ m \\ h \end{pmatrix}$$

$$\frac{dn}{dt} = \alpha_n (1-n) - \beta_n n$$

$$\frac{dm}{dt} = \dots \dots \dots$$

$$\frac{dh}{dt} = \dots \dots \dots$$

$$L_e = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$d_e = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\bar{F} = 0$$

$$f = \begin{bmatrix} \alpha_n (1-n) - \beta_n n \\ \alpha_m (1-m) - \beta_m m \\ \alpha_h (1-h) - \beta_h h \end{bmatrix}$$

$$n_0 = \frac{\alpha_n(V_n)}{\alpha_n(V_n) + \beta_n(V_n)}$$

$$\frac{dn}{dt} = 0$$

$$m_0 = \dots\dots\dots,$$

$$\frac{dm}{dt} = 0$$

$$h_0 = \dots\dots\dots$$

$$\frac{dh}{dt} = 0$$